Summary Test Maths

This covers some remaining parts from sections of Algebra (A4) from the edexcel specification.

Topics tested (Use Maths Genie to learn if needed):

1. Expanding and Factorising (Q1 - Q5)
2. Expanding and Factorising Quadratics (Q6 - Q8)
3. Factorising Harder Quadratics (Q9 - Q13)
4. Surds (Q14 - Q19)

Q1.

(a) Expand $7(2x + 7)$

(b) Factorise $3y + 12$

Q2.

(a) Factorise fully $12m + 8m^2$

(b) Solve $3(n - 5) = 27$

Q3.

(a) Factorise fully $5a^2b + 15ab^2$

(b) Solve $6(c - 8) = 42$

Q4.

(a) Factorise fully $30x^3 + 12x$

Q5.

(a) Factorise fully $18a^2bc + 30abc^2$

Q6.

(a) Expand and simplify $(2x + 3)(3x - 1)$

(b) Factorise $x^2 + 10x + 25$

Q7. (Hint part (b) - difference of 2 squares)

(a) Expand and simplify $(5h + 2)(h + 4)$

(b) Factorise $x^2 - 49$

Q8.

(a) Expand and simplify $(7x + 1)(x + 5)$

(b) Factorise $y^2 + 13y + 30$

Q9.

Factorise $2x^2 + 5x + 2$

Q10.

Solve $4x^2 - 19x - 5 = 0$

Q11.

Factorise fully $2x^2 - 98$

Q12.

Factorise $9x^2 - 3x - 20$

Q13.

Factorise $2x^2 + 13xy + 15y^2$

Q14.

Write $\sqrt{48}$ in the form $k\sqrt{3}$, where k is an integer.

Q15.

Expand and Simplify $(2 + \sqrt{3})(2 - \sqrt{3})$

Q16.

Write $(3 - \sqrt{2})^2$ in the form $a + b\sqrt{2}$, where a and b are integers.

Q17.

Rationalise the denominator $\frac{6}{\sqrt{3}}$

Q18.

Simplify $\frac{(3 + \sqrt{6})}{\sqrt{3}}$

Q19.

Show that $\frac{3\sqrt{3} + 3}{3 + \sqrt{3}}$ can be written as $\sqrt{3}$

Answers:

A1.

(a) Expand $7(2x + 7)$

$14x + 49$

(1)

(b) Factorise $3y + 12$

$3(y + 4)$

A2.

(a) Factorise fully $12m + 8m^2$

$4m(3 + 2m)$

(2)

(b) Solve $3(n - 5) = 27$

$3n - 15 = 27$

$3n = 42$

$n =$ 14

A3.

(a) Factorise fully $5a^2b + 15ab^2$

$5ab(a + 3b)$

(2)

(b) Solve $6(c - 8) = 42$

$6c - 48 = 42$

$6c = 90$

$c = 15$

$c =$ 15

(2)

A4.

(a) Factorise fully $30x^3 + 12x$

$6x(5x^2 + 2)$

A5.

(a) Factorise fully $18a^2bc + 30abc^2$

$6abc(3a + 5c)$

A6.

(a) Expand and simplify $(2x + 3)(3x - 1)$

$6x^2 - 2x + 9x - 3$

$6x^2 + 7x - 3$

(2)

(b) Factorise $x^2 + 10x + 25$

25

1 25

5 5

$(x + 5)(x + 5)$

A7.

(a) Expand and simplify $(5h + 2)(h + 4)$

$5h^2 + 20h + 2h + 8$

$5h^2 + 22h + 8$

(2)

(b) Factorise $x^2 - 49$

$(x + 7)(x - 7)$

(1)

A8.

(a) Expand and simplify $(7x + 1)(x + 5)$

$7x^2 + 35x + x + 5$

$7x^2 + 36x + 5$

(2)

(b) Factorise $y^2 + 13y + 30$

1	30
2	15
3	10
5	6

$(y + 3)(y + 10)$

(2)

A9.

Factorise $\overset{A}{2}x^2 + 5x + \overset{C}{2}$

$A \times C = 4$

1×4

2×2

$$\frac{(2x + 1)(2x + 4)}{2}$$

$(2x + 1)(x + 2)$

$(2x+1)(x+2)$

A10.

Solve $4x^2 - 19x - 5 = 0$

$AC = 20$

1	20
2	10
4	5

$$\frac{(4x+1)(4x-20)}{4}$$

$$(4x+1)(x-5) = 0$$

$$x = -\frac{1}{4} \quad x = 5$$

$$x = -\frac{1}{4} \quad x = 5$$

A11.

Factorise fully $2x^2 - 98$

$$2(x^2 - 49)$$

$$2(x+7)(x-7)$$

$$2(x+7)(x-7)$$

A12.

Factorise $9x^2 - 3x - 20$

$$\frac{(9x+12)(9x-15)}{9}$$

$$(3x+4)(3x-5)$$

180	
1	180
2	90
3	60
4	45
5	36
6	30
9	20
10	18
12	15

$(3x+4)(3x-5)$

A13.

Factorise $2x^2 + 13xy + 15y^2$

$$\frac{(2x+3y)(2x+10y)}{2}$$

$$(2x+3y)(x+5y)$$

30	
1	30
2	15
3	10
5	6

$(2x+3y)(x+5y)$

A14.

Write $\sqrt{48}$ in the form $k\sqrt{3}$, where k is an integer.

$\sqrt{16}\sqrt{3}$

$4\sqrt{3}$

$4\sqrt{3}$

A15.

Expand and Simplify $(2+\sqrt{3})(2-\sqrt{3})$

$4 - 2\sqrt{3} + 2\sqrt{3} - 3$

1

1

A16.

Write $(3-\sqrt{2})^2$ in the form $a + b\sqrt{2}$, where a and b are integers.

$(3-\sqrt{2})(3-\sqrt{2})$

$9 - 3\sqrt{2} - 3\sqrt{2} + 2$

$11 - 6\sqrt{2}$

$11 - 6\sqrt{2}$

A17.

Rationalise the denominator $\frac{6}{\sqrt{3}} \frac{\times\sqrt{3}}{\times\sqrt{3}}$

$\frac{6\sqrt{3}}{3}$

$2\sqrt{3}$

$2\sqrt{3}$

A18.

Simplify $\frac{(3+\sqrt{6})}{\sqrt{3}} \begin{matrix} \times\sqrt{3} \\ \times\sqrt{3} \end{matrix}$

$\sqrt{18} = \sqrt{9}\sqrt{2}$
$= 3\sqrt{2}$

$$\frac{3\sqrt{3} + \sqrt{18}}{3}$$

$$\frac{3\sqrt{3} + 3\sqrt{2}}{3}$$

$$\cancel{\frac{6\sqrt{2}}{3}} \qquad \underline{\underline{\sqrt{3} + \sqrt{2}}}$$

A19.

Show that $\frac{3\sqrt{3}+3}{3+\sqrt{3}}$ can be written as $\sqrt{3}$

$$\frac{(3\sqrt{3}+3)(3-\sqrt{3})}{(3+\sqrt{3})(3-\sqrt{3})}$$

$$\frac{9\sqrt{3} - 3(3) + 9 - 3\sqrt{3}}{9 - 3\sqrt{3} + 3\sqrt{3} - 3}$$

$$\frac{6\sqrt{3}}{6}$$

$$\underline{\underline{\sqrt{3}}}$$

AQA GCSE Combined Science – Chemistry
Topic 5.1: Atomic Structure and the Periodic Table
Summary Test

1. What is an atom? (1 mark)

2. What is an element? (1 mark)

3. What is a compound? (1 mark)

4. Complete the table of subatomic particles. (3 marks)

Proton – charge? relative mass?

Neutron – charge? relative mass?

Electron – charge? relative mass?

5. Which particle has a negative charge? (1 mark)

A) Proton B) Neutron C) Electron

6. Name the separation technique used to separate: (3 marks)

a) Sand from water

b) Water from salt water

c) Different dyes in ink

7. What is a mixture? (1 mark)

8. Explain the difference between a compound and a mixture. (2 marks)

9. Describe the plum pudding model of the atom. (2 marks)

10. The gold foil experiment led to a new model of the atom.

a) Who carried out this experiment? (1 mark)

b) What did the experiment show about the structure of the atom? (2 marks)

11. What is an isotope? (2 marks)

12. Write the electron configuration for: (3 marks)

a) Oxygen

b) Sodium

c) Calcium

13. Describe how you would obtain pure copper sulfate crystals from copper sulfate solution. (3 marks)

14. Explain why elements in the same group have similar chemical properties. (2 marks)

15. An atom has atomic number 12 and mass number 24. Complete: (3 marks)

Protons = ? Neutrons = ? Electrons = ?

16. Describe Rutherford's alpha scattering experiment and explain how the results led to the nuclear model. (4 marks)

17. Explain why fractional distillation can separate liquids in a mixture. (3 marks)

18. In chromatography a dye travels 4 cm and the solvent travels 8 cm. Calculate the Rf value. (2 marks)

19. Explain why isotopes of the same element have the same chemical properties. (2 marks)

20. Describe how the atomic model developed from Dalton to the modern model. (6 marks)

Mark Scheme

1. Smallest part of an element that can exist.

2. Substance made of only one type of atom.

3. Two or more different elements chemically bonded together.

4. Proton: +1, mass 1 | Neutron: 0, mass 1 | Electron: –1, mass 1/1836.

5. C) Electron.

6. a) Filtration b) Simple distillation c) Chromatography.

7. Two or more substances not chemically bonded together.

8. Compound: elements chemically bonded, new properties. Mixture: not chemically bonded and can be separated physically.

9. Atom is a sphere of positive charge with electrons embedded throughout.

10a. Ernest Rutherford.

10b. Atom mostly empty space; small dense positively charged nucleus.

11. Atoms of the same element with the same number of protons but different numbers of neutrons.

12. Oxygen: 2,6 | Sodium: 2,8,1 | Calcium: 2,8,8,2.

13. Heat solution to evaporate some solvent, allow to cool, crystals form and can be filtered/dried.

14. Same number of outer shell electrons.

15. Protons = 12, Electrons = 12, Neutrons = 12.

16. Alpha particles fired at thin gold foil; most passed through, some deflected, few bounced back; conclusion: atom mostly empty space with small dense nucleus.

17. Liquids have different boiling points; lowest boiling point evaporates first and condenses separately.

18. Rf = 4 ÷ 8 = 0.5.

19. Same number of electrons and same electronic structure.

20. Dalton: solid sphere → Thomson: electron & plum pudding → Rutherford: nucleus (gold foil) → Bohr: electron shells → Chadwick: neutron.

Revision in flashcards

Topic 1 - Atomic structure and the periodic table

5.1 Atomic structure and the periodic table
5.1.1 A simple model of the atom, symbols, relative atomic mass, electronic charge and isotopes

5.1.1.1 Atoms, elements and compounds

Q1. What is an atom?
A1. An atom is the smallest part of an element. They are the building blocks of all substances.

Q2. What is an element?
A2. A substance made of only one type of atom. There are about 100 elements and can be found in the periodic table.

Q3. How can elements be represented?
A3. They can be represented via a chemical symbol. For example the chemical symbol (Na) represents **one** atom of the element Sodium.

Q4. What is a compound?
A4. A substance formed when 2 or more different elements chemically bond together. For example water is 2 hydrogen and one oxygen atom chemically bonded together.

Q5. Compounds can also be represented with chemical symbols by combining the elements that make up the compound. If water is 2 hydrogen and one oxygen atom what would its chemical symbol be?
A5. H_2O

Q6. What do chemical reactions involve?
A6. The formation of one or more new substances, and often a detectable energy change.

Q7. How can compounds be separated into their elements?
A7. Through chemical reactions.

Q8. How do compounds mainly differ from the elements that make them up?
A8. Compounds have different unique properties from the elements they contain. E.g - the element sodium is a reactive metal and chlorine is a toxic gas but its compound sodium chloride is just table salt.

5.1.1.2 Mixtures

Q9. What is a mixture?
A9. Two or more elements or compounds **not** chemically combined together.

Q10. How is a mixture different from a compound?
A10.

- The chemical properties of each substance in the mixture are unchanged (keep original properties).
- A mixture can be separated with physical processes, unlike compounds that can only be separated via chemical reactions.

Q11. What is a physical process in the context of separating mixtures?
A11. A method of separation that does not involve chemical reactions and no new substances are made.

Q12. Name the physical process that can be used to separate mixtures.
A12.

- Filtration
- Crystallisation
- Simple distillation
- Fractional distillation
- Chromatography

Q13. Briefly describe and explain each physical process used for separating mixtures.
A13.
Filtration - Separates insoluble (a solid that doesn't dissolve in a liquid) solid from a liquid e.g sand + water.

- Mixture poured through filter paper in a funnel.
- Liquid passes through.
- Solid stays on the filter paper.

Crystallisation - Separates a soluble (solid that does dissolve in liquid) solid from a liquid e.g salt from salt water.

- Heat solution to evaporate the solvent (usually the liquid that dissolved the soluble solid).
- This leaves behind the separated solid in a crystal form. (you have to be careful that heating the liquid doesn't effect the thing we are trying to crystalise)

Simple distillation - Separates a liquid from a solid if we want to keep the liquid e.g water from salt water. (crystalisation does the same thing but the liquid is lost through evaporation)

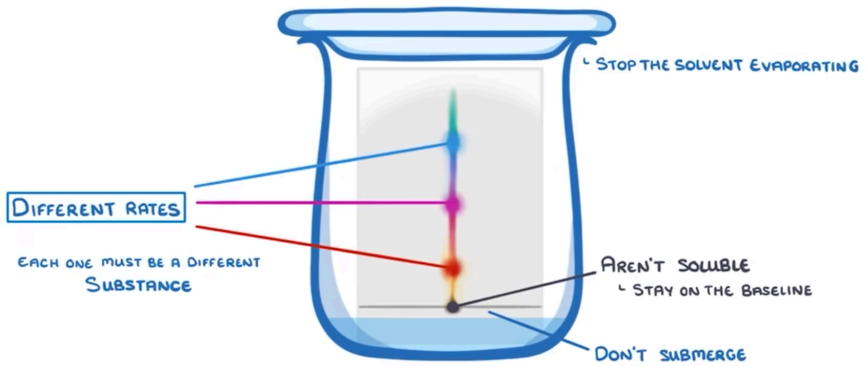
- Evaporate the liquid by heating.
- Vapour travels into a condenser cooling it.
- Vapour cools and becomes liquid (distillate)

Fractional distillation - Separate mixtures of liquids (they MUST have different boiling points for this method to work) e.g - ethanol and water.

- Mixture heated.
- The liquid with the lowest boiling point evaporates.
- The vapour travels up the fractionating column.
- The column allows vapours to cool and condense repeatedly, helping separate liquids with similar boiling points.
- It condenses and is collected as a liquid.

Chromatography (specifically **paper** chromatography) - Separates mixtures of soluble substances such as inks and food colorings based on their different solubilities in a solvent.

- Take a piece of chromatography paper and draw a line in **pencil** near the bottom. This is called the baseline.
- Add your sample of ink you suspect to contain a mixture of substances to the pencil line.
- Get your solvent (water or ethanol).
- Add the chromatography paper to the solvent making sure you don't submerge it more than the pencil line.
- Wait giving the solvent time to go up the paper dissolving your sample.
- Leave your paper to dry.
- You are left with a chromatogram.



Q14. Explain how paper chromatography works?

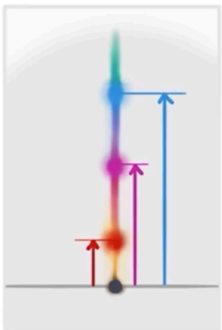
A14.

- As the solvent goes up the paper the different substances that make up our sample will dissolve in the solvent at different rates.
- Our substances dissolve in the solvent. Move a little bit (this is called the mobile phase since they are moving) and then, once again binding to the paper and staying still (called the stationary phase). Then they will again move a little bit, then stop moving again and so on. This happens over and over.
- Since more soluble substances will spend more time in the mobile phase than less soluble ones which will spend more time in the stationary phase. This makes the more soluble substance move 'faster' and higher up the paper.
- This separates the different substances since they have different solubilities hence spend more or less time in the mobile phase.

- If there are any insoluble substances in the sample they will not dissolve and won't move up the paper staying at the baseline.

Q15. How can we use the chromatogram to identify substances?
A15.

- How far each substance moves up the paper is dependent on the properties of the substance. Hence will be the same everytime anyone does the experiment.
- So you can measure the length of how much a substance moves up the paper and compare it to a data book which has the lengths and corresponding substance and use that to identify your substance.



Q16. What is a better way to identify the substance in chromatography than just looking at the length?
A17.

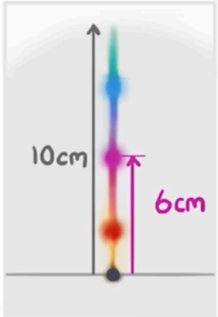
- How far the substance moves up the paper is dependent on how long you leave the experiment for. (Leaving it longer means the substance moves up more).
- So instead of comparing just the length we calculate the ratio called the R_f value.

Q18. How is the R_f value calculated?
A18.

$$\text{'}R_f \text{ VALUE'} = \frac{\text{DISTANCE TRAVELLED BY THE SUBSTANCE}}{\text{DISTANCE TRAVELLED BY THE SOLVENT}}$$

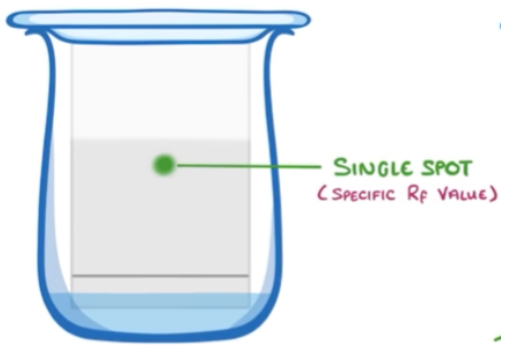
Q19. Why is the R_f value more reliable?
A19.
It is more reliable since it's not affected by time.
If the experiment is left for longer, indeed the substance will move higher up the paper giving us a different length but so will the solvent. Hence giving us the same ratio (R_f value) no matter how long we left the experiment for.
You can the use a data book of R_f values and compare to identify your substance.

Q20. If a substance moves up 6cm from the baseline and the solvent moves up 10cm. What is the R_f value?

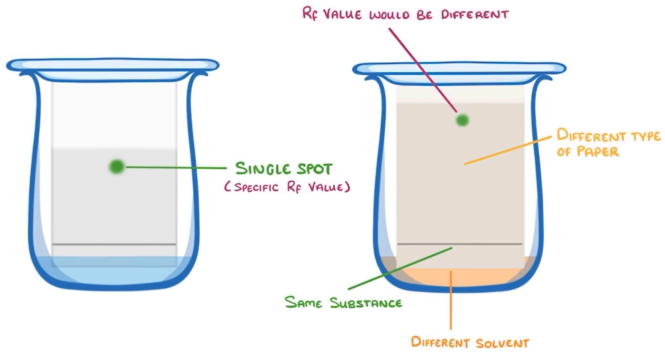


A20. $R_f = 6 / 10 = 0.6$

Q21. What will happen if we use paper chromatography on a pure substance that has nothing else mixed into it ?
A21. If the substance is soluble it will still move up the paper but won't separate out. Leaving a single spot on the paper. You can calculate a specific R_f value for this.



Q22. What happens if you put the same pure substance but in a different solvent or use a different type of paper.
A22. Since the substance might be more or less soluble with that solvent or more or less clinging to the paper. It will spend less or more time in the mobile and stationary phase. So you will still get a single spot since there is nothing to separate but it will be in a different place along the paper.



Because of this when looking up R_f values you also need to look up the solvent and paper used.

5.1.1.3 The development of the model of the atom

Q23. What was the initial idea of what atoms where and who proposed it?
A23. John Dalton - thought atoms to be tiny spheres that could not be divided.

Q24. What did JJ Thomson discover and how did this change our understanding of the atom?
A24. He discovered electrons, which lead to the plum pudding model.

Q25. What is the plum pudding model?
A25.

- Atom = sphere of positive charge.
- Electrons embedded throughout.
- It was called plum pudding because electrons were like plums in a pudding.

Q26. What experiment and by who lead to the disproving of the plum pudding model.
A26. Rutherford's - Alpha particle scattering experiment (also called gold foil experiment) led to the plum pudding model being disproved.

Q27. Describe the alpha particle scattering experiment.
A27.

- A beam of alpha particles was aimed at a very thin golf foil piece.
- The passage of the alpha particles was detected using a fluorescent screen.
- It was found that most alpha particles passed straight through. Some were deflected slightly at different angles. An even smaller number of them bounced back.

Q28. What were the findings of the experiment and why did the alpha particles behave as they did?
A29.

- Alpha particles have a net positive charge. Hence would be repelled / deflected if it came across another positive charge source.
- Since most alpha particles weren't deflected. This suggests that most of the atom is empty space.
- Hence the ones that were deflected or bounced back must be due to a small concentration of positive charge (and mass) in the atom (This is what we now call the nucleus which has protons giving it a positive net charge hence causing the alpha particles to deflect).
- The electrons orbit the nucleus in some way.
- This was called the **nuclear model** which replaced the plum pudding model.

Q29. What was Niels Bohr's contribution to the model?
A29.

- He suggested that electrons orbit at specific distances from the nucleus.
- These are called energy levels (shells).
- Electrons cannot exist between shells.
- This was the improved nuclear model and was accepted since the theoretical calculations of Bohr agreed with experimental observations.

Q30. What were the last 2 discoveries from the improved nuclear model that lead to the Modern Atomic model we have now?
A30.

- Protons and neutrons.
- Later experiments showed that the positive charge of any nucleus could be subdivided into a whole number of smaller particles, each particle having the same amount of positive charge (protons).
- James Chadwick's work: provided the evidence to show the existence of neutrons within the nucleus (This was about 20 years after the nucleus became an accepted scientific idea).

Q31. What is the summary of the development of the atomic model?
A31.
Dalton - Atoms are solid spheres.
Thomson - Electron discovered (plum pudding model).
Rutherford - alpha particle scattering experiment leads to nucleus discovery (nuclear model).
Bohr - Electron shells.
More experiments and Chadwick - Proton and neutron discovered (Modern atomic model).

5.1.1.4 Relative electrical charges of subatomic particles

Q32. Define atomic number.
A32. The number of protons in an atom of an element.

Q33. What is the relative charge of a proton, neutron and electron.
A33.

- Proton = +1
- Neutron = 0
- Electron = -1

Q34. What is the overall charge of an atom 0?
A34.

- The number of electrons in an atom are equal to the number of protons in the nucleus.
- For each positive 1 proton there is a -1 electron to cancel out that charge.
- Neutrons don't contribute to charge since they have 0.
- This makes the overall charge of an atom 0.

Q35. What determines which element an atom is?
A35.

- All atoms of a particular element have the same number of protons.
- Atoms of different elements have different numbers of protons.

5.1.1.5 Size and mass of atoms

Q36. What is the relative mass of a proton, neutron and electron?
A36.

- Proton = 1
- Neutron = 1
- Electron = very small / negligible

Q37. Where is most of the mass of an atom?
A37. In the nucleus of the atom since that's where the protons and neutrons are stored and electrons don't contribute much to the atom's mass.

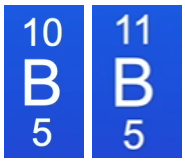
Q38. What is the mass number?
Q39. The sum of the protons and neutrons in a **single** atom,

Q39. The mass number of sodium (Na) is 23 and the atomic number is 11. How can this be represented?
A39.

(Mass number) 23
(Atomic number) 11 Na

Q40. Define an isotope.
A40. Atoms of the same element can have different numbers of neutrons; these atoms are called isotopes of that element.

Q41. Here are 2 examples of isotopes of Boron. What is the amout of protons and neutrons in each?



A41.

- They both have 5 protons, since the atomic number is 5 and all atoms of an element must have the same atomic number.
- The first one has a mass number (protons + neutrons) of 10 hence 5 neutrons and the second must have 6 neutrons.

Q42. What does carbon-14 mean?
A42. It shows that this is an isotope of carbon with mass number 14.

Q43. What is the radius of an atom and its nucleus roughly?
A43.

- Atom = 0.1 nm (1 x 10-10 m)
- Radius = less than 1/10 000 of that of the atom (about 1 x 10-14 m).
-

Q44. An atom of magnesium has: Atomic number = 12 and Mass number = 24. How many protons, neutrons and electrons does the atom have? Explain your answer.
A44.
Protons = 12
Neutrons = 12
Electrons = 12
The atomic number (12) is the number of protons by definition. Electron number is the same as proton number for a neutral charge atom. Mass number is the sum of protons and neutrons in an atom. So the neutron number is just - mass number - atomic number.

5.1.1.6 Relative atomic mass

Q45.What is the mass of a **single** atom?
A45. Protons and neutrons have a mass of 1 and electrons don't have much mass. Hence the mass of a single atom is just its mass number (number of protons and neutrons in the atom).

Q46. What is the relative atomic mass of an element?
A46. The relative atomic mass of an element is an average value that takes account of the abundance of the isotopes of the element.

Q47. Why do we have a **relative** atomic mass rather than just a mass value which would be the mass number of the atoms of that element?
A47.

- Reminder - isotopes are atoms of the same element but with different numbers of neutrons.
- Most elements on our Earth exist as a mixture of isotopes.
- Hence the same element has atoms with a different mass number due to its isotopes.
- We need to take into account ALL the atoms for an element. Hence take into account all the different isotopes / mass numbers and get an average.

Q48. What does abundance of an isotope mean?
A48.

- It is the percentage amount of that isotope on Earth.
- So if it said that carbon-14 has an abundance of 20%. It would mean 20% of all natural carbon on Earth is the isotope carbon-14 or you could say 20% of all carbon has a mass number of 14.

Q49. How do you calculate the relative atomic mass of carbon? (in this example we will assume carbon has only 2 isotopes carbon-14 and carbon-12)
A49.

- Look at all the isotopes of the element in this case carbon-14 and carbon-12.
- We need to take an average of both the mass numbers to get a relative atomic mass value.
- We can't just get the mean of 14 and 12 we need to take into account the **abundance** (amount) of each isotope present.
- Carbon-12 has an abundance of 80% and carbon-14 has abundance 20%.
- So we need to take carbon-12 more into account - mass number x abundance = 12 x 80 = 960.
- Carbon-14 - 14 x 20 = 280.
- We then add this up - 960 + 280 = 1240.
- And divide by the total abundance (this if all isotopes are taken into account should be 100% since all isotopes together make 100% of that element on Earth) - 1240 / 100 = 12.40.
- This is the **relative atomic mass of carbon and is what is shown on the periodic table**. (not exactly since we assumed carbon only has 2 isotopes).
- This might seem like a lot but can be summarized by the formula in Q50.

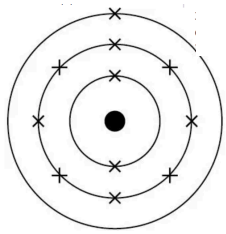
Q50. What is the formula for relative atomic mass of an element?
A50.

$$A_r = \frac{(\%\ of\ isotope\ a \times mass\ of\ isotope\ a) + (\%\ of\ isotope\ b \times mass\ of\ isotope\ b)}{100}$$

Q51. An element Y has 3 isotopes. Y-24 with abundance 79%, Y-25 with abundance 10% and Y-26 with abundance 11%. Calculate the relative atomic mass (Ar) of element Y. Give your answer to 2dp.
A51. 24.32

5.1.1.7 Electronic structure

Q52. Draw and talk about the general structure of electrons in an atom.
A52.
Electrons move around the nucleus in energy levels, also called electron shells.



Q53. Each shell can only have a certain max number of electrons. What are the maxes for the first 3 shells?
A53.

- First shell - 2 electrons
- Second shell - 8 electrons
- Third shell - 8 electrons (This is GCSE specific it's actually 18 but since we only look at the first 20 elements GCSE says 8)

Q54. The electronic structure of sodium is 2,8,1 what does this mean?
A54. Sodium has 11 electrons, with 2 in the first shell, 8 in the second and 1 in the third.

5.1.2 The periodic table

5.1.2.1 The periodic table

Q55. What is the periodic table?
A55. A table of all known elements arranged in a way that shows patterns in their properties.

Q56. Briefly explain how the periodic table is arranged?
A56.

- Elements are arranged in order of increasing atomic number (number of protons).
- Elements with similar properties are in columns, known as groups.
- Elements in the same group have the same number of electrons in their outer shell (outer electrons). This gives those elements similar chemical properties.

5.1.2.2 Development of the periodic table

Q57. How did scientists first try to classify elements?
A57. They arranged them in order of their atomic weight. This was since protons, electrons and neutrons were not discovered yet.

Q58. What were the problems with this method of using atomic weight and early periodic tables in general?
A58.

- Led to some elements being placed in inappropriate groups.
- Tables were incomplete since many elements hadn't been discovered yet.

Q59. How did Mendeleev overcome some of these problems?
A59.

- Left gaps for elements that he thought had not been discovered.
- In some places he changed the order based on atomic weight.
- Then later elements with properties predicted by Mendeleev were discovered and filled the gaps.

Q60. What discovery explained why the order based on atomic weight was not always correct?
A60. Isotopes.

Q61. Explain how testing a prediction can support or refute a new scientific idea.
Q61.

- Scientists make predictions based on a scientific idea.
- If experiments match the prediction, the idea is supported.
- If results do not match, the idea may be changed or rejected.
- When new elements were discovered, their properties matched Mendeleev's predictions, which supported his periodic table.

Q62. So what led to the modern periodic table?
A62.

- Elements with properties predicted by Mendeleev were discovered and filled in gaps.
- Knowledge of isotopes explained why order based on atomic weight wasn't always correct.
- Discovery of electrons, protons and neutrons lead to the elements being ordered based on their atomic number.
- This leads to the table fixing itself with elements with similar properties being in the same groups.

5.1.2.3 Metals and non-metals

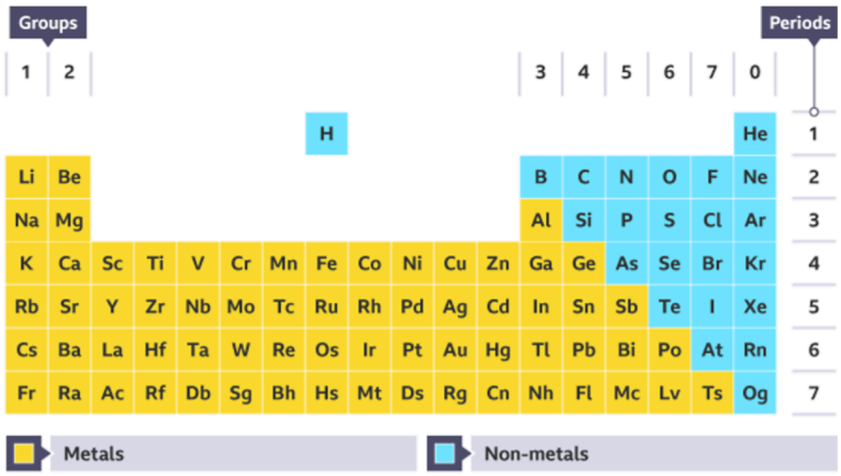
Q63. What is an ion?
A63.

- Atoms usually have 0 net charge.
- This is because the number of electrons (-1 charge) is equal to the number of protons (+1 charge). Hence the charges cancel out giving an overall net 0 charge.
- An ion is an atom or group of atoms that has gained or lost one or more electrons.
- This creates an imbalance in the number of protons and electrons giving the atom an overall net positive (if atom lost electron) or negative (if atom gained electron) charge.

Q64. What is the difference between metals and non-metals?
A64. Elements that react and lose electrons to form positive ions are metals. Hence elements that react and don't form positive ions are non-metals.

Q65. How are metals and non metals displayed in the periodic table?
A65.

- The majority of elements are metals.
- Metals are found to the left and towards the bottom of the periodic table.
- Non-metals are found towards the right and top of the periodic table.



Q66. What are some common characteristics of metals?
A66.

- Good conductors of electricity
- Good conductors of heat
- Strong and dense
- Malleable (can be hammered into shape)
- Usually have high melting and boiling points

Q67. What are some common characteristics of non-metals?
A67.

- Poor conductors of electricity
- Poor conductors of heat
- Low melting and boiling points
- Often gases or brittle solids
- Not malleable

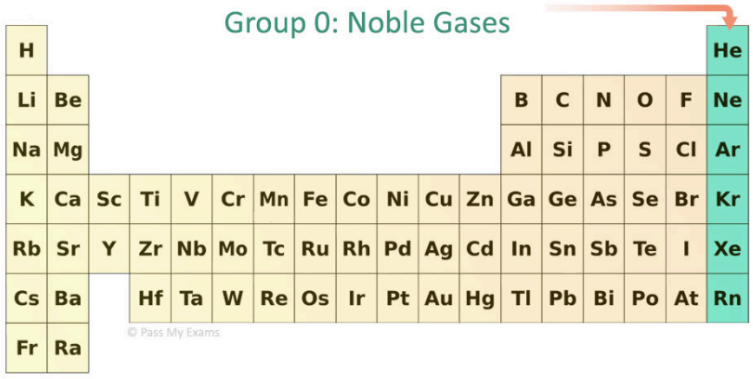
Q68. Explain how the atomic structure of metals and non-metals relates to their position in the periodic table.
A68.

- Elements are arranged in the periodic table by increasing atomic number (number of protons).
- The atomic number determines the electron arrangement in an atom.
- Elements on the left side (metals) have few electrons in their outer shell.
- Elements on the right side (non-metals) have many electrons in their outer shell.

Q69. Explain how the reactions of elements are related to the arrangement of electrons in their atoms and hence to their atomic number.
A69.

- The atomic number determines the number of electrons in an atom.
- Electrons are arranged in shells around the nucleus.
- The outer shell electrons control how an element reacts (since those require less energy to escape the atom).
- Elements in the same group have the same number of outer electrons, so they have similar chemical reactions.

5.1.2.4 Group 0



Q70. What are the elements in group 0 called?
A70. The noble gases.

Q71. How are the electrons arranged in most of the noble gases?
A71.
They have eight electrons in their outer shell, except for helium, which has only two electrons. They all have a full outer electron shell.

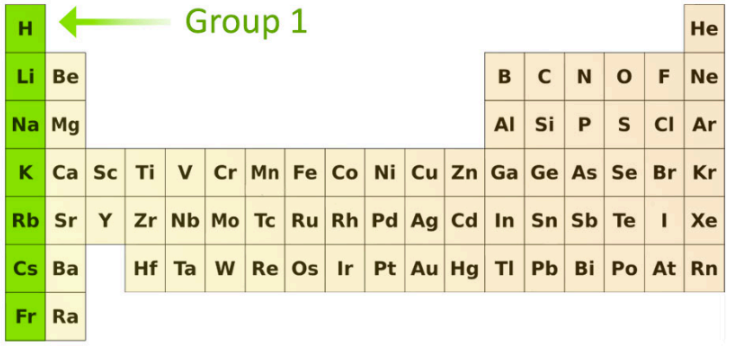
Q72. Why are the noble gases very unreactive?
A72 .

- They have a full electron outer shell.
- A full outer shell is very stable, so these elements do not easily gain or lose electrons.
- This is why noble gases are very unreactive.

Q73. What are some trends seen in Group 0?
A73. As you go down Group 0:

- Boiling points increase
- Melting points increase
- This is because the atoms become larger (higher relative atomic mass), so more energy / heat is needed to boil them.

5.1.2.5 Group 1



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Q74. What are the elements in group 1 called?
A74. The Alkali metals (minus hydrogen).

Q75. What is the electron structure of the alkali metals?
A75. They all have one electron in their outer shell (including hydrogen).

Q76. What is the reactivity of the Alkali metals (so no hydrogen) like and why?
A76.

- Group 1 metals are very reactive.
- This is because they only have one electron in their outer shell. Having a fuller outer shell makes the atom more stable which isn't the case here. So losing a single electron is very easy.
- They all react vigorously with water to create an alkaline solution (hence the name) and hydrogen.
- They all react with oxygen to create an oxide.
- They all react with chlorine to form a white precipitate.

Q78. How does reactivity change going down the group and why?
A78.

- Reactivity increases going down the group.
- This is because as you go down the outer electrons are further from the nucleus.
- This means the nucleus pulls on the outer electrons with less force making them easier to lose.

Q79. What are some of the properties of group 1 metals (i.e not including hydrogen)?
A79.

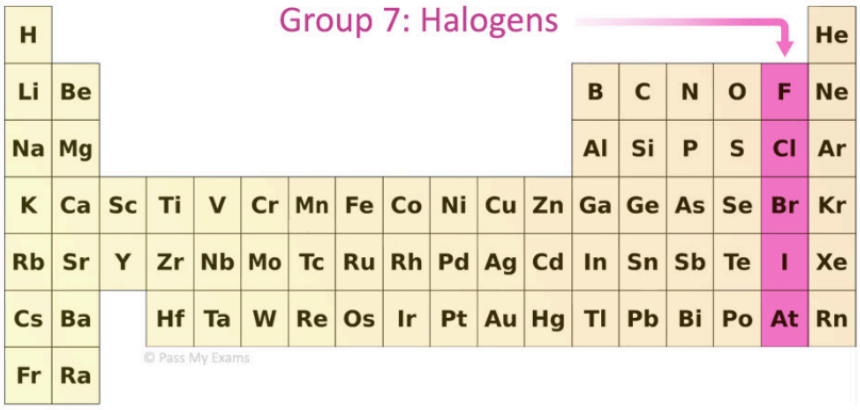
- Soft metals.
- Have relatively low melting points.
- Are good conductors of heat and electricity.
- Very reactive.

Q80. Describe the reactions of the first three alkali metals with oxygen, chlorine and water.
A80.

- This is Lithium (Li), Sodium (Na) and Potassium (K). Hydrogen is in group 1 but not counted as an alkali metal but a gas.
- If asked for Group 1 elements, hydrogen may be included.
- If asked for alkali metals, start from lithium.

	reaction with oxygen	reaction with water	reaction with chlorine
lithium	burns with a strongly red-tinged flame and produces a white solid	Fizzes steadily, gradually disappears	white powder is produced and settles on the sides of the container
sodium	strong orange flame and produces white solid	Fizzes rapidly, melts into a ball and disappears quickly	burns with a bright yellow flame, clouds of white powder are produced and settle on the sides of the container
potassium	large pieces produce lilac flame, smaller ones make solid immediately	Ignites with sparks and a lilac flame, disappears very quickly	reaction is even more vigorous than with sodium

5.1.2.6 Group 7



Q81. What are the elements in group 7 known as?
A81. Halogens.

Q82. What is the electron structure of halogens?
A82. Have 7 electrons in their outer shell. This means they have similar reactions.

Q83. Are halogens metals or non metals?
A83, Non-metals

Q84. What is a molecule?
A84. A substance made of more than one atom held together by covalent bonds.

Q85. How do halogens naturally exist?
A85. They naturally exist as a molecule. This means you can't naturally have a single Bromine atom (Br) but instead a molecule of 2 bromine atoms (Br_2).

Q86. What happens as you go down group 7?
A86.

- The relative mass, melting point, boiling point all increase.
- Halogens in general are very reactive but reactivity of the elements decreases going down the group.

Q87, Why does the reactivity decrease as you go down the group?
A87.

- Atoms become larger.
- The outer shell is further from the nucleus. (This would usually make the atom more reactive since it's easier to lose an electron. However halogens typically react by gaining an electron. Since the nucleus is deeper in the atom as you go down the group it's harder for it to attract an electron.)

Q88. How do halogens react with metals?
A88.

- Halogens react with metals to form ionic compounds.
- Halogen atoms gain one electron.
- Metals lose electrons.
- This forms ionic compounds called metal halides.
- E.g - Sodium + Chlorine → Sodium chloride (NaCl).

Q89. How do halogens react with non-metals?
A89.

- Halogens react with non-metals to form covalent compounds.
- Non-metals share electrons with halogens.
- This forms covalent bonds.
- E.g - Hydrogen + Chlorine → Hydrogen chloride (HCl).

Q90. What can happen when halogens meet?
A90.

- A more reactive halogen (higher up in the group) can displace a less reactive halogen from an aqueous solution of its salt.
- E.g - Chlorine will displace bromine if we bubble the gas through a solution of potassium bromide.
- Chlorine + potassium bromide → potassium chloride + bromine.
- This happens because chlorine is more reactive than bromine.